

1	(a)	150 or 1.5×10^2 Gm	A1	[1]
	(b)	distance = $2 \times (42.3 - 6.38) \times 10^6$ (= 7.184×10^7 m) (time =) $7.184 \times 10^7 / (3.0 \times 10^8) = 0.24$ (0.239) s	C1 A1	[2]
	(c)	units of pressure P : $\text{kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$ units of density ρ : kg m^{-3} and speed v : m s^{-1} simplification for units of C : $C = v^2 \rho / P$ units: $(\text{m}^2 \text{s}^{-2} \text{kg m}^{-3}) / \text{kg m}^{-1} \text{s}^{-2}$ and cancelling to give no units for C	M1 M1 A1	[3]
	(d)	energy and power (<i>both underlined and no others</i>)	A1	[1]
	(e)	(i) vector triangle of correct orientation three arrows for the velocities in the correct directions	M1 A1	[2]
		(ii) length measured from scale diagram 5.2 ± 0.2 cm or components of boat speed determined parallel and perpendicular to river flow velocity = 2.6 m s^{-1} (allow $\pm 0.1 \text{ m s}^{-1}$)	C1 A1	[2]